



Computer Organization and Assembly language

Course: Computer Organization and Assembly Language (Lab) - CS-318

1. Conversions

1. How do you convert a decimal number to hexadecimal manually?
2. How do you convert a binary number to decimal?
3. How do you convert a hexadecimal number to binary?
4. What is the difference between BCD and binary representation?
5. How can ASCII characters be converted to numeric values in assembly?
6. How do you store a decimal number in a register?
7. What instruction is used for arithmetic conversion in 8086?
8. How do you convert a number from lower case to upper case using ASCII values?
9. Explain signed vs unsigned conversion.
10. How do you represent negative numbers in 8086?
11. How do you convert an ASCII digit to its integer value in 8086?
12. How is hexadecimal used in memory addressing?
13. What is the use of AAA instruction?
14. Explain DAA instruction for BCD adjustment.
15. How do you handle overflow during conversion?

2. Installation

1. How do you install EMU8086 on Windows?
2. What are the system requirements for EMU8086?
3. How do you configure DOSBox for assembly programming?
4. What is the difference between MASM and EMU8086?
5. How do you create a new assembly program in EMU8086?

6. What file extension is used for 8086 assembly programs?
7. How do you assemble and run a program in EMU8086?
8. What is the difference between assemble and compile in EMU8086?
9. How do you enable keyboard interrupts in EMU8086?
10. How do you debug a program in EMU8086?
11. How do you set memory segment size in EMU8086?
12. What is the difference between real mode and protected mode in DOS?
13. How do you save an assembled program in EMU8086?
14. How do you open an existing program in EMU8086?
15. How do you view registers and memory contents during execution?

3. Registers

1. Name all general-purpose registers in 8086.
2. What is the difference between 8-bit and 16-bit registers?
3. What is the purpose of the AX register?
4. Explain the function of the BX register.
5. What is the purpose of the CX register?
6. What is the purpose of the DX register?
7. Name the segment registers and their use.
8. What is the difference between SP and BP?
9. What is the function of SI and DI registers?
10. Explain the role of the FLAGS register.
11. Which register is used for string operations?
12. How is the IP register used during execution?
13. Which registers can be used for pointer arithmetic?
14. How is the AX register split into AH and AL?
15. What is the importance of the CS register?

4. Arithmetic Instructions

1. Name the basic arithmetic instructions in 8086.
2. How does the ADD instruction work?
3. How does the SUB instruction work?
4. What is the difference between INC and ADD?

5. Explain MUL and IMUL instructions.
6. How does DIV work?
7. How do you handle carry using ADC?
8. How do you handle borrow using SBB?
9. Which flag registers are affected by arithmetic operations?
10. How is NEG instruction used?
11. How can you perform subtraction using addition?
12. Explain the difference between signed and unsigned multiplication.
13. How is overflow detected in arithmetic operations?
14. How do you add two 16-bit numbers using 8-bit registers?
15. How do you perform BCD addition?

5. Addressing Modes

1. What is immediate addressing? Give an example.
2. What is register addressing? Give an example.
3. What is direct addressing?
4. Explain indirect addressing mode.
5. What is register indirect addressing?
6. What is based addressing mode?
7. What is indexed addressing mode?
8. What is based plus index addressing mode?
9. How do you access memory using SI and DI?
10. Which addressing mode is fastest?
11. How is segment:offset used in addressing?
12. Explain relative addressing.
13. How do you calculate effective address in 8086?
14. Which addressing mode is used for arrays?
15. Can immediate data be used in memory operations?

6. Display a Message

1. Which DOS interrupt is used to display a character?
2. How do you display a string using INT 21h?
3. Which register stores the ASCII code for a character?

4. How do you display a newline in DOS?
5. Which function of INT 21h is used for printing a string?
6. How do you terminate a program after displaying a message?
7. How do you store a string in memory?
8. How do you print a message character by character?
9. Can you use loops to display a string?
10. How do you handle \$ in string printing with DOS?
11. How do you display numeric values as characters?
12. Which register holds the segment of a string for printing?
13. How do you print multiple messages sequentially?
14. How can you display a message in uppercase only?
15. How is a message terminated in memory for DOS printing?

7. Logic Operators

1. Name the basic logic instructions in 8086.
2. How does AND work in 8086?
3. How does OR work?
4. How does XOR work?
5. How is NOT used?
6. Which flags are affected by logic operations?
7. How can you use AND to mask bits?
8. How can you toggle bits using XOR?
9. How can you check a specific bit using AND?
10. How is SHL used in logic operations?
11. How is SHR used in logic operations?
12. How do you perform bitwise inversion?
13. How do you combine multiple logic operations?
14. How can you set or reset a bit in a register?
15. How is logical comparison done using CMP?

8. Arrays

1. How do you declare an array in 8086?
2. How do you access array elements?

3. Which registers are used for array indexing?
4. How do you loop through an array?
5. How is SI used for array traversal?
6. How is DI used for destination indexing?
7. How do you sum all elements of an array?
8. How do you find the largest element of an array?
9. How do you store array elements in memory?
10. Can arrays be multidimensional?
11. How do you reverse an array in 8086?
12. How do you copy one array to another?
13. How do you initialize an array with constants?
14. Which addressing mode is used for arrays?
15. How do you handle array overflow?

9. Loops

1. Which instructions are used for loops in 8086?
2. How does LOOP instruction work?
3. How do you implement a counting loop?
4. How do you implement an infinite loop?
5. How do you implement a loop with JCXZ?
6. Which register controls the number of iterations?
7. How do you nest loops in 8086?
8. How can loops be used to traverse arrays?
9. How do you decrement a counter in a loop?
10. How do you terminate a loop on a condition?
11. How do you use CMP inside a loop?
12. How do you use JMP to create a loop?
13. How do loops affect the CX register?
14. How do you implement a sum of numbers using a loop?
15. How do you implement a factorial using a loop?

10. Jumps

1. What is the difference between conditional and unconditional jumps?

2. Which instruction is used for unconditional jump?
3. Name conditional jump instructions.
4. How does JE/JZ work?
5. How does JNE/JNZ work?
6. How does JC work?
7. How does JNC work?
8. How does JL/JNGE work?
9. How does JG/JNLE work?
10. How does JO/JNO work?
11. How do jumps work with loops?
12. How do you implement a simple decision using jump?
13. How do you implement a nested if using jumps?
14. Which flag registers are checked during jumps?
15. How do jumps affect program flow?

11. Bubble Sort

1. What is bubble sort algorithm?
2. How do you implement bubble sort in 8086?
3. Which registers are used as counters in bubble sort?
4. How do you compare two adjacent array elements?
5. How do you swap elements in 8086?
6. How do loops work in bubble sort?
7. How do you optimize bubble sort for fewer passes?
8. How do you detect if array is already sorted?
9. How do you store the sorted array back in memory?
10. Which addressing mode is used for array elements?
11. How do you handle multiple passes in bubble sort?
12. How is CX register used in outer loop?
13. How is inner loop implemented?
14. How do you increment pointers to traverse the array?
15. How do you handle array of bytes vs words?

12. Shifts

1. What is the difference between logical and arithmetic shift?
2. How does SHL work?
3. How does SHR work?
4. How does SAL differ from SHL?
5. How does SAR work for signed numbers?
6. How many bits can you shift in one instruction?
7. How do shifts affect carry flag?
8. How do you multiply a number by 2 using shift?
9. How do you divide a number by 2 using shift?
10. How do you shift multi-bit numbers in registers?
11. How do you implement bit masking using shift?
12. How is ROL different from SHL?
13. How is ROR different from SHR?
14. Can you shift memory contents directly?
15. How do shifts help in bit manipulation?

13. Rotation

1. What is rotation in 8086?
2. How does ROL work?
3. How does ROR work?
4. How does RCL differ from ROL?
5. How does RCR differ from ROR?
6. How do carry flags affect rotation?
7. How do you rotate multiple bits?
8. How do you use rotation for encryption?
9. How do you rotate a byte stored in memory?
10. How do you rotate 16-bit numbers?
11. How is rotation different from shift?
12. How do you rotate circularly?
13. How do you rotate signed numbers?
14. How do rotations affect high and low bits?

15. How do you use rotation in masking operations?

14. Subroutine

1. What is a subroutine in 8086?
2. How do you define a subroutine?
3. How do you call a subroutine?
4. How do you return from a subroutine?
5. How do you pass parameters to a subroutine?
6. Which register is used for return address?
7. How is the stack used in subroutines?
8. How do local variables work in subroutines?
9. Can a subroutine call another subroutine?
10. How do you implement recursion?
11. How do flags behave in subroutines?
12. How do you save registers before a subroutine call?
13. How do you restore registers after return?
14. How do you use subroutines for code reuse?
15. How do you handle multiple return points?

15. Subroutine with Parameters

1. How do you pass parameters through registers?
2. How do you pass parameters through stack?
3. How do you access parameters inside subroutine?
4. How do you return values from subroutine?
5. Which registers are commonly used for parameters?
6. How do you pass arrays as parameters?
7. How do you pass byte vs word parameters?
8. How do you increment pointer to access multiple parameters?
9. How do you implement subroutine with conditional execution?
10. How do you handle nested subroutine calls with parameters?
11. How do you preserve registers while using parameters?
12. How do you pass constants as parameters?
13. How do you pass memory addresses to subroutine?

14. How do you combine multiple subroutines with parameters?
15. How do you debug parameter passing in EMU8086?